

Data Acquisition



ONTOLOGY SEARCH

build keywords for the search query using Ontology Search.



retrieves disease papers to form corpus.

Feature engineering

Gene Embeddings

to extract a semantic profile for each gene



THE HUMAN PROTEIN ATLAS

integrate word embeddings with tissue expression from Human Protein Atlas (HPA)

Gene Prioritization

Logistic Regression

by combining text + biological features.



Gene ranking

across each locus related to the disease



Validation

Evaluation Metrics

AUC-ROC, Recall@k and validation with OpenTargets



Network analysis

to evaluate whether text embeddings converge to similar modules as biological-only models

